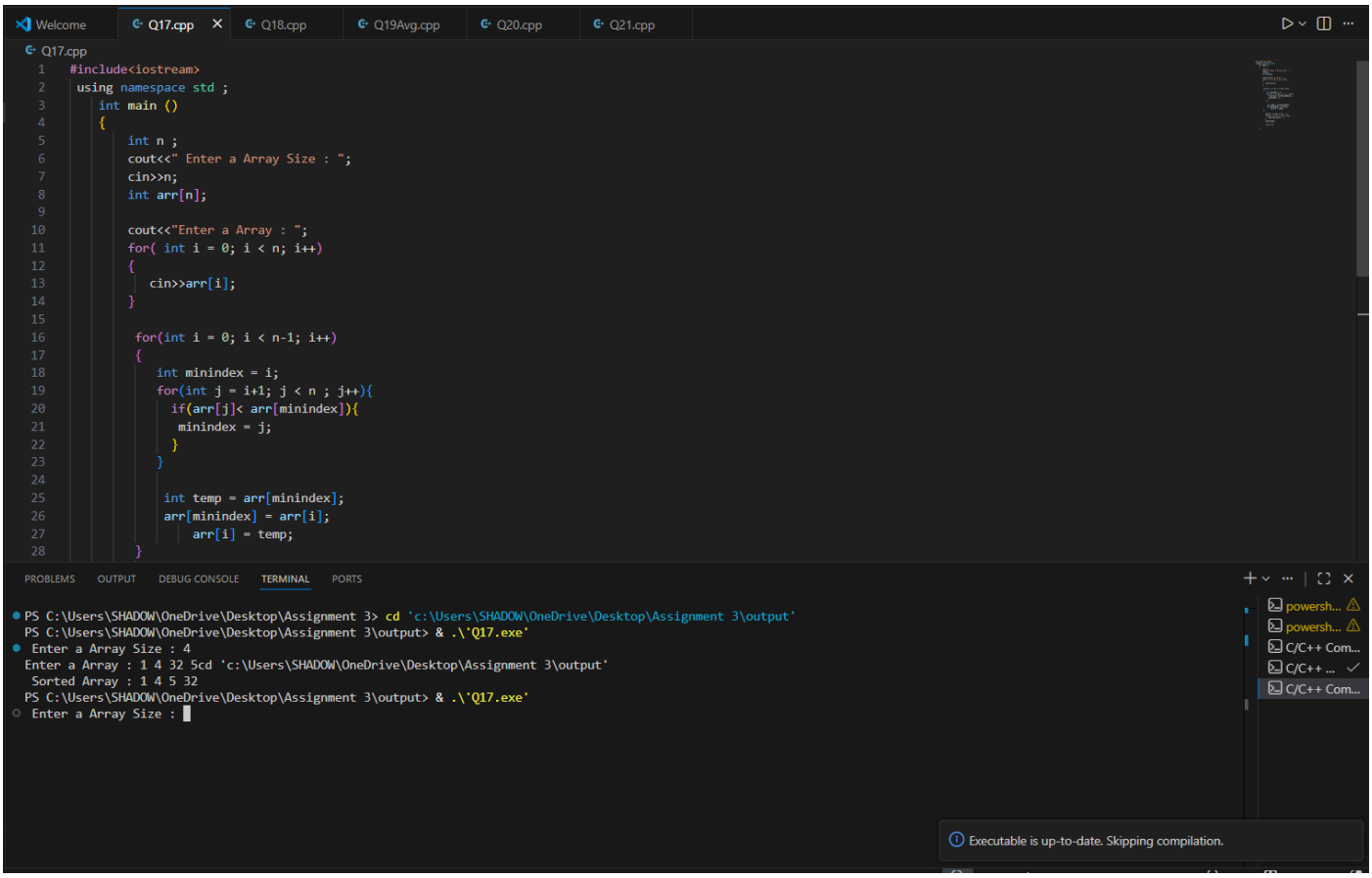
17. Accept n numbers from users and store them in to array. Write a function to sort and display sorted array. i/p:[1,4,32,5] o/p:[1,4,5,32].

```
#include<iostream>

 using namespace std ;

   int main ()

   {

      int n ;

      cout<<" Enter a Array Size : ";

      cin>>n;

      int arr[n];


      cout<<"Enter a Array : ";

      for( int i = 0; i < n; i++)

      {

        cin>>arr[i];

      }


       for(int i = 0; i < n-1; i++)

       {

         int minindex = i;

         for(int j = i+1; j < n ; j++){

           if(arr[j]< arr[minindex]){

            minindex = j;

           }

         }


           int temp = arr[minindex];

           arr[minindex] = arr[i];

              arr[i] = temp;

       }


        cout<<" Sorted Array : ";

        for(int i = 0; i < n; i++){

           cout<<arr[i]<<" ";
```

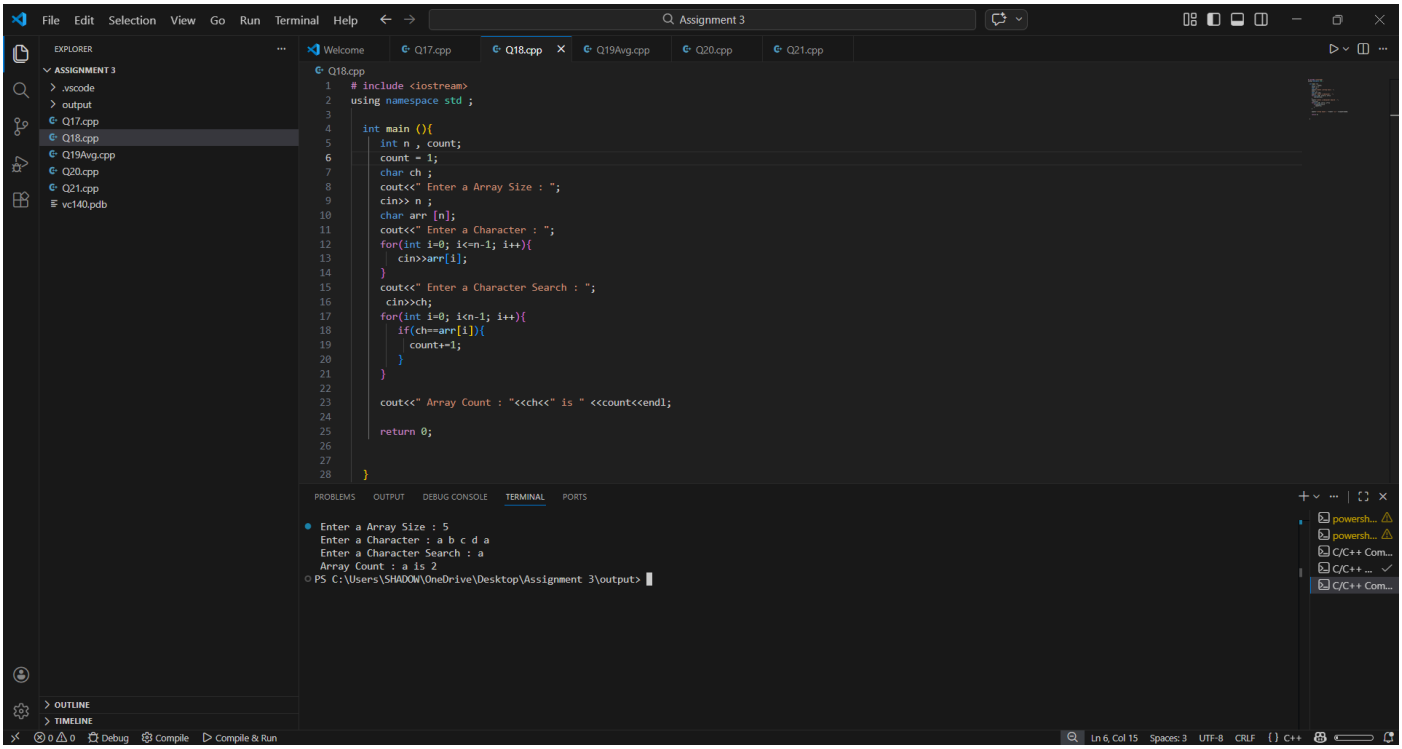
```
    }
    cout<<endl;


    return 0;


}
```



18. Accept n characters from user and search a character appears how many times in the array. i/p:['a','b','c','d','a'] search 'a' o/p: a count is 2.

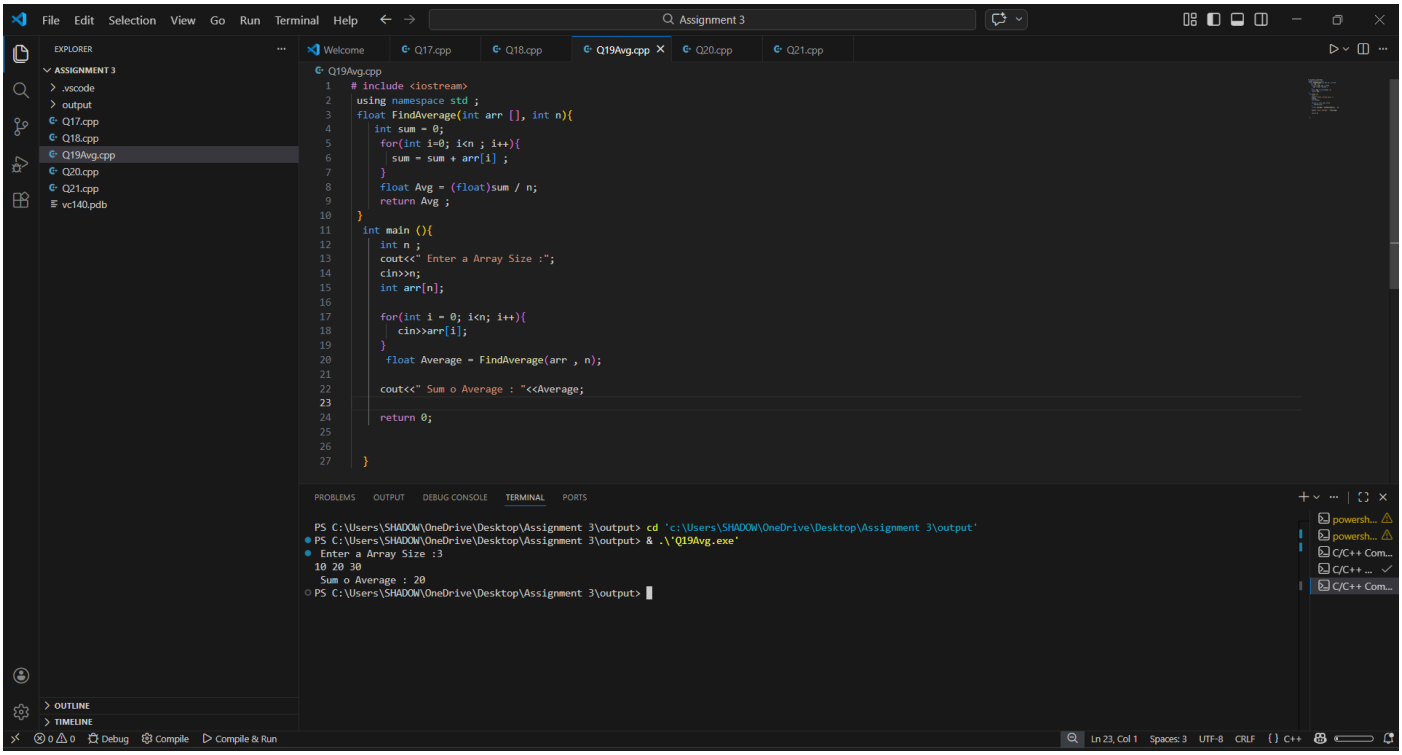
```
# include <iostream>

using namespace std ;


 int main (){

   int n , count;

   count = 0;

   char ch ;

   cout<<" Enter a Array Size : ";

   cin>> n ;

   char arr [n];

   cout<<" Enter a Character : ";
```

```
for(int i=0; i<=n-1; i++){
  cin>>arr[i];
}
cout<<" Enter a Character Search : ";
 cin>>ch;
for(int i=0; i<n-1; i++){
  if(ch==arr[i]){
   count+=1;
  }
}

cout<<" Array Count : "<<ch<<" is " <<count<<endl;

return 0;

}
```



19. Accept n numbers from user and write a function to find out average and display average.
i/p:{10,20,30} o/p:Average: 20

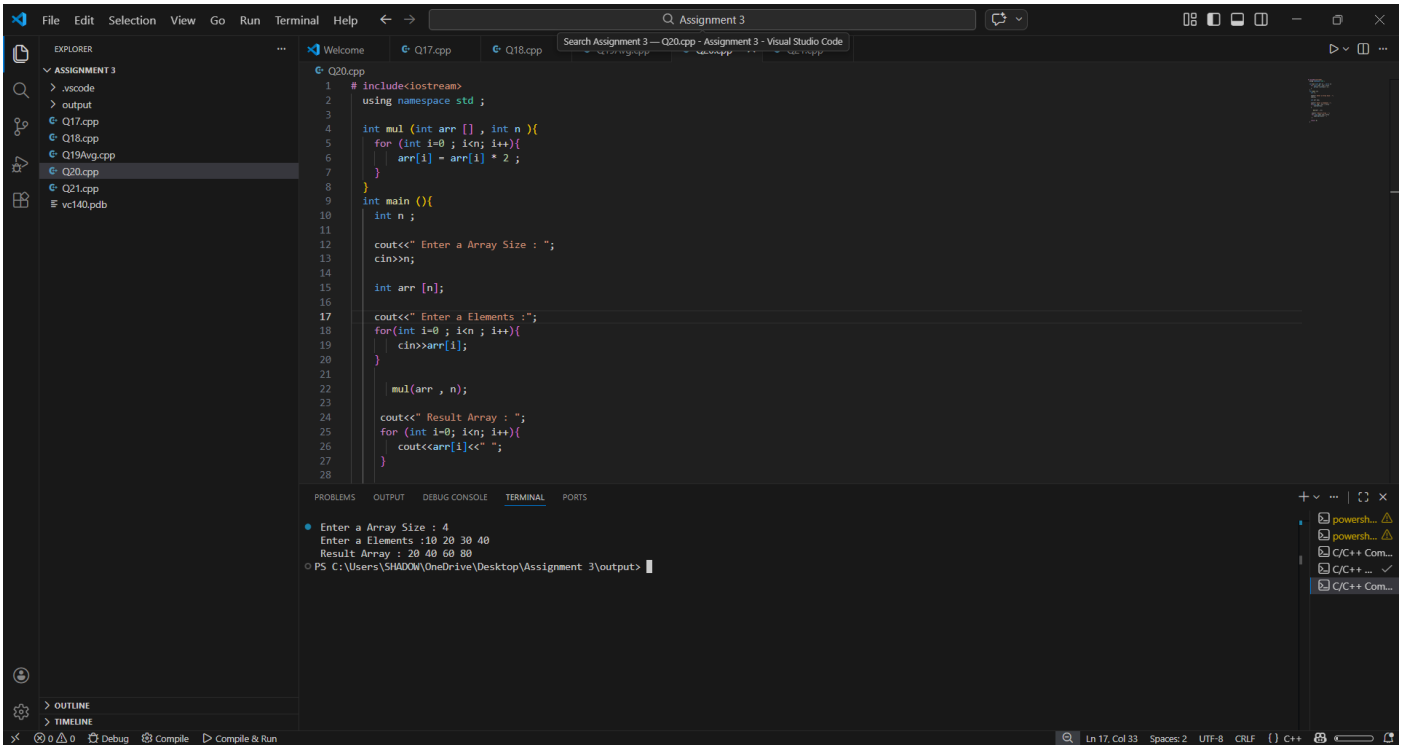
```
# include <iostream>

 using namespace std ;

 float FindAverage(int arr [], int n){

   int sum = 0;

    for(int i=0; i<n ; i++){

      sum = sum + arr[i] ;

    }

    float Avg = (float)sum / n;

    return Avg ;

 }

  int main (){

    int n ;

    cout<<" Enter a Array Size :";

    cin>>n;

    int arr[n];


    for(int i = 0; i<n; i++){

      cin>>arr[i];

    }

     float Average = FindAverage(arr , n);


    cout<<" Sum o Average : "<<Average;


    return 0;


  }
```



20. result. Accept n numbers from user and write a function to multiply each num-ber by 2 and display i/p: [10,20,30,40] o/p:[20,40,60,40]

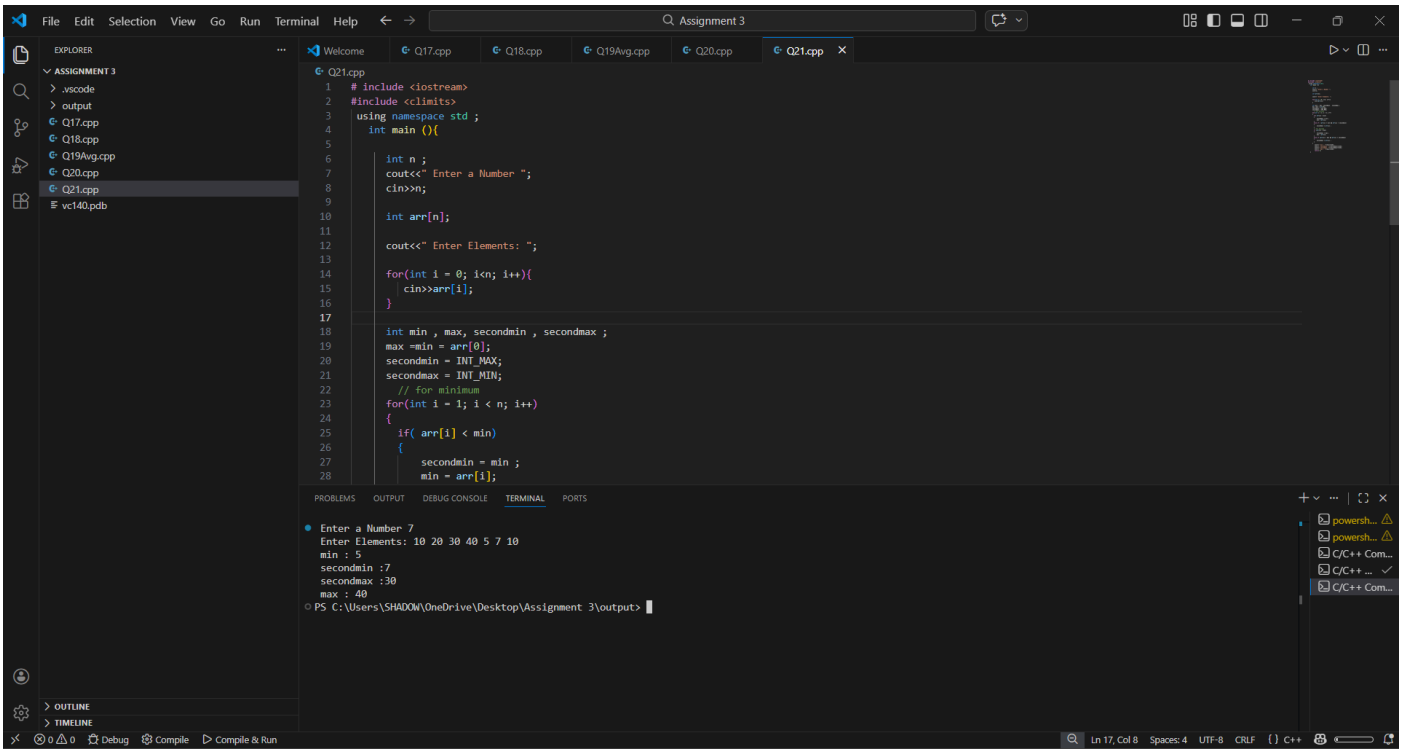
```
# include<iostream>
 using namespace std ;


 int mul (int arr [] , int n ){
  for (int i=0 ; i<n; i++){
     arr[i] = arr[i] * 2 ;
  }
 }
 int main (){
  int n ;


  cout<<" Enter a Array Size : ";
  cin>>n;


  int arr [n];


  cout<<" Enter a Elements :";
  for(int i=0 ; i<n ; i++){
     cin>>arr[i];
```

```
  }

   mul(arr , n);

  cout<<" Result Array : ";
  for (int i=0; i<n; i++){
    cout<<arr[i]<<" ";
  }

 return 0;
}
```



21. Accept n number from user and write functions for finding min, max, second max and second min from these numbers i/p: [10,20,30,40,5,7,10] o/p:min:10 second min: 20 max:40 second max:30

```
# include <iostream>

#include <climits>

 using namespace std ;

  int main (){


    int n ;

    cout<<" Enter a Number ";

    cin>>n;
```

```
int arr[n];

cout<<" Enter Elements: ";

for(int i = 0; i<n; i++){
  cin>>arr[i];
}

int min , max, secondmin , secondmax ;
max =min = arr[0];
secondmin = INT_MAX;
secondmax = INT_MIN;
 // for minimum
for(int i = 1; i < n; i++)
{
 if( arr[i] < min)
 {
    secondmin = min ;
    min = arr[i];
 }
 else if ( arr[i] > min && arr[i] < secondmin)
 {
    secondmin = arr[i] ;
 }
 // for maximum
 if (arr[i] > max)
 {
    secondmax = max ;
    max = arr[i];
 }
 else if (arr[i] < max && arr[i] > secondmax)
 {
    secondmax = arr[i] ;
```

```
    }
  }
    cout<<" min : "<<min<<endl;
    cout<<" secondmin :"<<secondmin<<endl;
    cout<<" secondmax :"<<secondmax<<endl;
    cout<<" max : "<<max<<endl;
    return 0;
}
```